Gradient-Based Temporal Explanations for Speech Emotion Recognition

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1 Introduction

Speech emotion recognition models make utterance-level decisions from a signal that mixes lexical content, prosody, pauses, speaker variation, and background conditions. Pastor et al.'s SpeechXAI work addresses this interpretability problem with an intuitive perturbation protocol: align words in the audio, silence each word-related segment, and measure how much the target-class probability changes [1]. Their explanation unit is therefore a spoken word, which is easy to inspect and discuss.

Our project asks a narrower research question: **do model-specific gradient-based temporal attributions identify more decision-relevant regions than Pastor/SpeechXAI word-level attributions?** We also test what is gained or lost by moving from word granularity to smaller HuBERT token intervals across two different emotional-speech conditions: conversational IEMOCAP and controlled RAVDESS. The comparison is not between different classifiers. SpeechXAI/Pastor, Level 3, and LeGrad-HuBERT are all evaluated on the same HuBERT model and the same masking operation.

2 Related Work

Pastor et al. position SpeechXAI against the broader XAI literature in vision, language, and tabular data, including gradient saliency, Integrated Gradients, Grad-CAM, LIME, SHAP, and occlusion-style perturbation methods [1]. For speech, they emphasize that many previous explanations are time-frequency heatmaps or phoneme/fixed-frame analyses, which may be less accessible for non-expert users or less suitable for semantically intensive speech classification tasks. Their method follows an input-perturbation view: the relevance of an audio segment is the probability difference between the original utterance and the utterance with that segment masked.

The empirical setting uses two standard acted-emotion speech corpora with different acoustic and interaction profiles. IEMOCAP contains dyadic scripted and improvised conversations, so the model receives longer, more naturalistic utterances with speaker interaction, pauses, and variable prosody [3]. RAVDESS contains controlled North American English recordings of fixed sentences spoken by actors, providing a cleaner benchmark with less conversational variability [4]. Both datasets are evaluated with the same SUPERB HuBERT emotion-recognition checkpoint: HuBERT converts the waveform into temporal Transformer tokens, and the downstream SUPERB head maps those tokens to the emotion classes [7, 8]. The four-class label protocol is also part of the selected checkpoint's task definition: the model card states that SUPERB ER follows the conventional IEMOCAP protocol, dropping unbalanced emotion classes to leave four classes [9]. Therefore, differences in the results should be read as differences in explanation behavior across datasets, not as differences caused by training or changing the classifier.

Our work keeps Pastor's faithfulness question but changes the explanation source. Level 3 is a Transformer attention-flow attribution inspired by attention rollout [6] and class-specific head relevance. LeGrad-HuBERT adapts the LeGrad idea, which treats positive gradients with respect to attention maps as the explanation signal, originally proposed for Vision Transformers [5]. In Pastor et al.'s concept-based XAI taxonomy [2], SpeechXAI's word segments are close to *post-hoc, local, symbolic class-concept relations*: each spoken word is a human-readable concept whose effect on a class is measured. Our methods are more fine-grained and model-specific: they are *post-hoc, local, gradient-based temporal feature attributions* for Transformer audio tokens. They become concept-level explanations only if the selected token intervals are later grouped into words or higher-level speech concepts.

3 Research Gaps

Pastor/SpeechXAI gives explanations that are easy to read, but the word is a coarse unit. A word can contain several acoustic events, and emotion classifiers may rely on pauses, emphasis, hesitation, pitch, or short stressed regions that are not aligned with whole words. Conversely, token-level gradient methods can find smaller regions, but those regions are less immediately meaningful to humans. This creates the main gap studied here: **the trade-off between interpretability by word-level concepts and faithfulness by finer temporal granularity.**

A second gap is architectural. LeGrad was designed for Vision Transformers, where patch tokens remain in a visual-token pathway. The HuBERT sequence classifier used here has a different head:

$$\text{waveform} \rightarrow \text{HuBERT hidden states } H_0, \ldots, H_{12} \rightarrow H_{\text{mix}} \rightarrow \text{projector} \rightarrow \text{masked temporal mean pooling} \rightarrow \text{classifier.}$$

When weighted layer aggregation is active, the model computes

$$H_{\text{mix}} = \sum_{l=0}^{12} \text{softmax}(\alpha)_l H_l.$$

The stored α_l values are forward mixture parameters, not class-specific relevance scores. They say how much the trained head mixes each hidden-state tensor into H_{mix}, while LeGrad needs a scalar class score whose gradient can flow back to an attention map. This is why we evaluate explicit HuBERT adaptations rather than claiming a direct copy of the original ViT method.

4 Methodology

Model and target. The implemented pipeline loads the HuBERT checkpoint `superb/hubert-base-superb-er`, with 12 Transformer layers, 12 heads, 768 hidden dimensions, a 256-dimensional projector, masked temporal mean pooling, and four emotion labels: neutral, happy/excited, angry, and sad [7, 8, 9]. We align datasets to this trained output space: IEMOCAP excited is mapped to happy and unsupported labels are excluded, while RAVDESS is filtered to the exact matching emotions. Every explanation targets the model's original predicted class $\hat{y}$, not necessarily the dataset ground-truth label. This is important because the explanation is about the model decision being evaluated.

Pastor/SpeechXAI baseline. We reproduce the word-level SpeechXAI protocol as the baseline method. WhisperX provides word timestamps, each word interval is silenced, and the score is

$$r(x_i) = p(\hat{y} \mid x) - p(\hat{y} \mid x \setminus x_i).$$

The implementation follows the SpeechXAI repository behavior by applying the word mask in the waveform domain: the effective interval starts 100 ms before the aligned word and ends 40 ms after it, clipped to the audio boundaries. This produces word scores, ranked word intervals, and a human-readable explanation.

Level 3 relevance. Level 3 is the strongest rollout-based method retained for the report, but it is not default raw-attention rollout. Default rollout composes attention probabilities; our code first builds class-conditioned matrices from positive gradient-times-attention. For each HuBERT layer l, attention probabilities A_l and their target-logit gradients are captured. The layer matrix is

$$G_l = \text{mean}_h \left[\text{ReLU} \left(\frac{\partial s_{\hat{y}}}{\partial A_{l,h}} \odot A_{l,h} \right) \right],$$

where h indexes heads. The matrices are augmented with the identity, row-normalized, and multiplied through the Transformer stack to form the joint rollout matrix R. Since HuBERT has no ViT-style class token, we average the final-token rows to get `rollout_score`. In parallel, the classifier head gives `head_relevance`: each token contribution is computed through the effective linear weight `classifier.weight` $\times$ `projector.weight` under masked temporal mean pooling. The final method is

$$\texttt{level3} = \text{normalize}(\texttt{rollout\_score} \odot \texttt{head\_relevance}).$$

Thus Level 3 uses rollout and gradient-times-attention; it is a hybrid Transformer relevance method, not LeGrad.

LeGrad-HuBERT variants. LeGrad-HuBERT keeps the LeGrad distinction that mattered most in our discussion: it uses positive attention gradients directly and does not use rollout:

$$L_l = \text{mean}_{h,q} \left[\text{ReLU} \left(\frac{\partial s}{\partial A_{l,h,q,:}} \right) \right].$$

The source/key token axis is kept as the temporal explanation axis. We evaluate the best LeGrad family in two layer-aggregation variants. In the *layer-local* score, each Transformer output follows

$$H_l \rightarrow \text{projector} \rightarrow \text{masked temporal mean pooling} \rightarrow \text{classifier} \rightarrow s_{l,\hat{y}}.$$

This gives a score tied to layer l, closer in spirit to LeGrad. We aggregate L_l either by a uniform mean over layers or by the renormalized HuBERT layer weights, where the H_0 coefficient is dropped because H_0 has no attention map and the remaining 12 coefficients are renormalized. Final-score variants are implemented in the repository, but the report focuses on layer-local variants because they were the strongest LeGrad-HuBERT results.

Deletion with the same time removed. For each audio and each $k \in \{1, 2, 3, 5\}$, Pastor/SpeechXAI top-k words define a total amount of audio time X to remove. HuBERT token relevance vectors are first mapped to equal-width intervals over the full waveform duration; Level 3 and LeGrad-HuBERT then select their top intervals until they remove the same amount of time X. All methods silence their selected intervals without changing waveform length. The main metric is the confidence drop for the original predicted class:

$$\Delta p = p(\hat{y} \mid x) - p(\hat{y} \mid x_{\text{masked}}).$$

Higher Δp means the selected region was more important for the model's original decision. A random deletion by silence masking baseline selects random bins with the same total removed time and runs 20 trials per audio.

5 Experiments and Analysis

Before the dataset comparison, we validated the pipeline on the same IEMOCAP utterance. SpeechXAI alignment and word masking run end-to-end; HuBERT exposes 12 attention layers and 201 temporal tokens for that audio; LeGrad-HuBERT returns layer relevance tensors of shape $(1, 12, 201)$; and the independent Level 3 reconstruction of normalize($\texttt{rollout\_score} \odot \texttt{head\_relevance}$) has maximum absolute difference 0.0 from the implementation.

We evaluate IEMOCAP and RAVDESS separately. The available IEMOCAP run is partial and must not be described as a full-dataset result: it keeps 5460 audios out of 5531 expected, with one incomplete audio after filtering. The available RAVDESS run keeps 671 audios out of 672 expected. Table 1 reports mean confidence drop; numbers in parentheses are the relative change compared with Pastor/SpeechXAI at the same dataset and k. The second line reports the percentage of paired audios for which the method's confidence drop is larger than Pastor/SpeechXAI.

Dataset	k	Pastor	Random deletion	Level 3	LeGrad HuBERT-w.	LeGrad mean
IEMOCAP	1	0.115 baseline	0.046 (-59.7%) beats 19.7%	0.128 (+12.0%) beats 55.3%	0.106 (-7.3%) beats 43.6%	0.105 (-8.7%) beats 42.9%
IEMOCAP	2	0.159 baseline	0.078 (-51.0%) beats 22.3%	0.181 (+13.8%) beats 56.0%	0.155 (-2.4%) beats 46.3%	0.153 (-3.4%) beats 45.2%
IEMOCAP	3	0.183 baseline	0.103 (-43.9%) beats 25.3%	0.215 (+17.2%) beats 56.4%	0.185 (+1.1%) beats 46.6%	0.183 (+0.1%) beats 46.2%
IEMOCAP	5	0.207 baseline	0.139 (-32.8%) beats 30.5%	0.256 (+23.7%) beats 57.1%	0.221 (+7.0%) beats 49.6%	0.220 (+6.4%) beats 49.5%
RAVDESS	1	0.109 baseline	0.027 (-75.6%) beats 3.6%	0.104 (-4.4%) beats 45.2%	0.161 (+47.7%) beats 72.4%	0.163 (+49.5%) beats 72.3%
RAVDESS	2	0.199 baseline	0.082 (-58.5%) beats 6.0%	0.216 (+9.0%) beats 55.7%	0.327 (+64.5%) beats 81.8%	0.328 (+65.0%) beats 80.8%
RAVDESS	3	0.249 baseline	0.138 (-44.5%) beats 15.6%	0.282 (+13.1%) beats 62.3%	0.413 (+65.8%) beats 84.2%	0.412 (+65.4%) beats 83.8%
RAVDESS	5	0.227 baseline	0.207 (-8.8%) beats 44.9%	0.342 (+50.6%) beats 76.2%	0.433 (+90.5%) beats 84.2%	0.429 (+88.9%) beats 83.5%

Table 1: Deletion results with the same amount of audio time removed, computed on the available duration-matched comparison runs: 5,460 IEMOCAP audios from a partial 5,531-audio run and 671 RAVDESS audios from 672 expected files. The first line in each cell is the mean confidence drop $\Delta p = p(\hat{y} \mid x) - p(\hat{y} \mid x_{\text{masked}})$; values in parentheses are the relative change versus Pastor/SpeechXAI for the same dataset and k. The second line reports the beat-Pastor rate: the percentage of paired audios for which the method's confidence drop is larger than Pastor/SpeechXAI. Pastor/SpeechXAI uses word intervals; Random deletion, Level 3, and LeGrad-HuBERT use the same removed-time budget. For random deletion, the paired comparison uses the mean over 20 trials per audio.

The current evidence suggests two different regimes. On IEMOCAP, Level 3 is the strongest method across all k, improving over Pastor/SpeechXAI by 12.0% to 23.7%. LeGrad layer-local variants become competitive only when more duration is removed, especially at $k = 3$ and $k = 5$. On RAVDESS, LeGrad layer-local variants dominate, with gains above 47% for every k, while Level 3 is weaker at $k = 1$ and improves as the removed duration grows. This supports the project hypothesis that finer temporal granularity can improve deletion faithfulness, but it also shows that the best gradient-based construction depends on the dataset and audio distribution.

Several factors may explain this dataset gap. First, IEMOCAP contains conversational speech with longer turns, interruptions, pauses, and lexical variation; word-level removal can therefore capture semantically meaningful events, and rollout-style context propagation may help distribute relevance across longer dependencies. RAVDESS uses shorter controlled sentences, where the lexical content is repeated and emotion is expressed more strongly through local acoustic cues such as intensity, pitch, and voice quality; direct LeGrad attention-gradient scores may align better with these localized cues. Second, the HuBERT classifier is not equally reliable across datasets: the RAVDESS diagnostic run has 228 correctly predicted utterances out of 672, so the evaluated subset is biased toward decisions the model already handles well. Third, the same removed-time budget can correspond to different semantic units: in IEMOCAP, a word-level interval may cover rich conversational content, while in RAVDESS the same duration can mostly cover short acoustic emphasis. These points are plausible explanations rather than isolated causal tests, but they clarify why a single attribution method should not be assumed to dominate across speech corpora.

RAVDESS diagnostic evaluation. We also ran additional diagnostics on the 228 RAVDESS utterances that the HuBERT classifier predicted correctly. Deletion AOPC averages the target-class probability drop after silencing the top relevance tokens at 5%, 10%, 20%, and 30% budgets. Class-specificity is the Pearson correlation between predicted-class and runner-up-class relevance maps, where lower is better. Concentration is the relevance mass contained in the top 10% tokens, where higher is more localized.

Diagnostic metric	Level 3	LeGrad HuBERT-w.	LeGrad mean
Deletion AOPC $\Delta p \uparrow$	0.152	0.209	0.209
Class-specificity Pearson $\rho \downarrow$	-0.242	0.486	0.489
Top-10% relevance mass $\uparrow$	0.327	0.137	0.137

Legend. Deletion AOPC is the average confidence drop after removing the top-ranked temporal regions at the four tested budgets, so higher values indicate stronger deletion faithfulness. Class-specificity Pearson ρ compares predicted-class and runner-up-class relevance maps, so lower values indicate more class-dependent explanations. Top-10% relevance mass measures how much relevance is concentrated in the highest-scoring 10% of tokens, so higher values indicate more localized explanations.

Main conclusion. Pastor/SpeechXAI remains the most interpretable explanation because words are directly understandable. Our best gradient-based methods are less semantic but often more faithful when every method removes the same amount of audio time. The strongest measured claim is therefore specific: gradient-based HuBERT token attribution can outperform word-level Pastor/SpeechXAI on confidence-drop faithfulness, especially for RAVDESS and for larger removal budgets. The additional diagnostics show the trade-off behind that result: LeGrad gives the strongest RAVDESS deletion AOPC, while Level 3 gives more class-specific and more concentrated relevance maps.

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